Aadya Goel

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EDUCATION

Carnegie Mellon University Aug 2025- Jul 2027

MS Artificial Intelligence Engineering - Biomedical Engineering (Research Track)

- ***Honor & Award: Biomedical Engineering Department Head's Fellowship.***
- Poster Presenter, 2025 Carnegie Mellon Forum on Biomedical Engineering: presented the development of a genetic analysis tool to an audience of over 872 attendees, highlighting its applications in understanding viral pathogenic mechanisms.
- Emerging Leaders Mentor (2025-present), mentoring undergraduate students through the program to foster leadership, communication, and community engagement skills.

GPA - 4.00 , Dean's Honors

Vellore Institute of Technology, India Jul 2021- Jul 2025

B. Tech Computer Science with specialization in Bioinformatics

Courses: Biology, Data Analytics in Bioinformatics, Biochemistry, Molecular Biology, Analytical Bioinformatics, Cell Biology and Genetics, Biological Databases, Artificial Intelligence, Engineering Calculus, Probability and Statistics, Object Oriented Programming, Data Structures and Algorithms, Database Management Systems, Operating Systems, Software Engineering

CGPA - 8.83, Class Rank - 15th out of 174 students (Top 8% percentile)

RESEARCH PROJECTS

University of Pitsburgh Medical Centre- Graduate Student Researcher Sept 2025- Present

- Collaborating with Dr. Lianghui Zhang (UPMC), Dr. Siyang Zheng (CMU), and researchers from UPMC, NIH, and Yale on multi-omic and health technology research addressing virus-induced endothelial dysfunction.
- Developing AI/ML and systems biology pipelines integrating transcriptomic, proteomic, and clinical datasets from both human and mouse models.
- Building multiscale mathematical models to elucidate pathogenic mechanisms underlying acute and post-COVID chronic comorbidities.
- Integrating de-identified clinical data and multi-omic analyses with in vitro, in vivo, and ex vivo experiments to identify biomarkers and patient sub-phenotypes associated with therapeutic response.
- Applying machine learning to large-scale biomedical data from prospective (NIH ACTIV4c, Pitt Long COVID) and retrospective (NIH ACTIV4a, ACTIV5, completed Pitt Long COVID) cohorts for discovery and validation.
- ***Skills:*** *Machine Learning · Data Science · Python · Deep Learning · Cell Ranger ATC · R programming · Multi-omic data · Single-cell analysis*

Cambridge Centre for International Research (CCIR)- Research Scholar May 2024- Jun 2025

- Worked under Dr. Shadi Ghiasi, University of Oxford, on Artificial Intelligence in Healthcare in Biomedical sensors and Signal processing.
- Authored a paper titled 'Epilepsy Seizure Detection based on EEG Signals using Machine Learning' for detecting epilepsy in patients aged 3 to 22 using EEG brain signals with deep learning and signal processing techniques. ***Published and Presented at the International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (ICIITCEE 2025)***.
- ***Honor & Award: Received an admit in the highly competitive CCIR Future Scholars Program (admitting only 5 students per course) with a CCIR STEM scholarship (awarded to student researchers who exhibit great passion and academic interest in their fields).***
- ***Nominated as a CCIR Spotlight Scholar for demonstrating an exceptional spark of curiosity and for a singularly impressive research project among the cohort, and invited to the YouTube podcast series- CCIR Student Research Scholars.***
- ***Skills:*** *Machine Learning · Data Science · Signal Processing · Healthcare data · Biomedical Engineering · Mental Health · Deep Learning*

Identification of Suicidal Behavior using Deep Learning Models- College Capstone Jul 2023- Apr 2025

- Utilization of neural networks and Natural Language Processing (NLP) techniques to identify patterns indicative of suicidal tendencies in online social media and brain f-MRIs data to detect suicidal risk and classify them as adolescent individuals at risk of self-harm or suicide.
- ***Skills:*** *Data processing, deep learning, natural language processing*

Enhancing Image Feature Matching Detection: ORB & HDBSCAN algorithm integration Sep 2023- May 2024

- Evaluated orb and hbdscan algorithms as a hybrid method on various scientific image datasets to analyze similarity/ manipulation within them by performing image feature matching.
- Performed a comparative analysis showcasing superior performance of ORB-HDBSCAN combination, particularly in Homography-RANSAC.
- ***Skills:*** *OpenCV · Python · Scikit-Learn · Image Processing · Feature Extraction · NumPy · Feature Matching · Computer Vision · Matplotlib*
- ***Published at the 15th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT), June 24-28, 2024 at IIT Mandi, India***

Global Academic Internship Program, NUS Singapore · Research Intern Dec 2022- Jun 2023

- Completed the GAIP program on 'Data Analytics using Deep Learning' and collaborated with a team to develop a model for "Emotion Analysis in Sports" with 98.22% accuracy.
- Worked on a research project with Dr. Tan from NUS, Singapore, on "Detection of Stock Market Manipulation using Deep Learning,"
- ***Skills:*** *Research Skills · Machine Learning · Deep Learning · Data Science*
- ***Under peer review at a Springer journal and available for pre-print on Research Square Research Gate, and OUCI.***

Hewlett Packard Enterprise, Singapore · Onsite Internship- Research Intern Dec 2022- Jan 2023

- Completed a Big Data Analytics program, focusing on advanced techniques and tools, migrated applications to Microsoft Azure and implemented analytics on UCI Datasets for the final project on machine learning.
- ***Skills:*** *Big Data Analytics · Microsoft Azure*

WORK EXPERIENCE

SETV Global · Healthcare technology Startup

Head of Genomics & Lead – Elderly & Women Care Division

Feb 2025- July 2025

- Led a team of 40 interns to refine and enhance Gene-omere, optimizing its genomic analysis for genetic abnormality detection, organism classification, and pathogen matching using advanced hybrid agent workflows, while collaborating with healthcare professionals at AIIMS Delhi & Telangana Govt to expand its role in genetic health risk assessment and personalized healthcare.
- Conducted an in-depth analysis of Human Metapneumovirus (HMPV) using Gene-omere in response to rising cases in China and India; findings submitted to WHO, NIH, CEPI, and Indian medical authorities, which are under consideration in vaccine development.
- Oversaw the marketing and deployment of SETV solutions in the Elderly and Care divisions, for all product deliverables to ensure accessibility and adoption, while leading a team of 10 marketing interns to drive strategic outreach.
- ***Skills:*** *Python · Computational Genology · Genome Sequencing · Bioinformatics · NCBI BLAST · Cancer Genomics · Biopython · Marketing*
- ***Presented Gene-omere at Startup Mahakumbh 2025 in New Delhi, showcasing its impact on genomic research and precision medicine to a global audience of over 50,000 business leaders from 50+ countries, investors, and innovators as part of the 'Startup India @ 2047 —Unfolding the Bharat Story' initiative.***

Head of Genomics

April 2024- Feb 2025

- Led a team of 25 to develop Gene-omere, a pioneering genomic analysis tool capable of detecting genetic abnormalities, classifying 75 organisms, and matching pathogens using a custom RNN model. Implemented interactive visualization for DNA/protein sequences and a module for simulating targeted therapies, impacting personalized medicine.
- ***Skills:*** *Python · Computational Genology · Genome Sequencing · Bioinformatics · NCBI BLAST · Cancer Genomics · Biopython*
- ***Presented Gene-omere at the 2024 Global AI Summit in Hyderabad, showcasing its potential to revolutionize genomic research and personalized treatment strategies to over 2,500 industry leaders and innovators.***

AI/ML Intern at SETV Global

April 2024- Sep 2024

- Developed AI solutions for chronic disease management at SETV's Bhanu Damodar Oncological Research Center (BDORC).
- Developed a breast cancer multi-class classification and prediction ML model using mammography images and recordings using CNNs.
- Developed an intelligent generative chatbot, ***Meddonna***, using NLP and AI for healthcare assistance, featuring location-based services and multilingual speech & text support.
- ***Skills:*** *Python · Data Analysis · Computer Vision · Deep Learning · Natural Language Processing (NLP) · Speech Recognition · pyttsx3 · Optical Character Recognition (OCR) · REST APIs*
- ***Paper on "Leveraging Artificial Neural Networks for Breast Cancer Detection and Prognosis," presented and published at the IEEE International Conference on Computational, Communication and Information Technology (ICCCIT 2025)***

Thermo Fisher- Virtual Internship- Genetic Sciences

April 2024

- Developed expertise in genetic analysis, biotechnology, and bioinformatics by interpreting data and applying laboratory techniques.
- Gained practical experience in scientific communication and research through industry-relevant project focused on genetic sciences and cutting-edge biotechnological applications.
- ***Skills:*** *Understanding Real-time PCR (qPCR), Data science and visualization, Laboratory methodologies, Problem-solving in genetics and life sciences*

European School of Leadership Development, United Nations- Remote Internship

Sep 2023- May 2024

- Developed a deep learning-based climate trend prediction scale with Mr. Yerkanat Abilpatta (Department of Geoinformatics), analyzing historical data and climate variables like temperature and precipitation to improve prediction accuracy.
- ***Skills:*** *Python · Data Processing · Research Skills · Deep Learning*

CodeClause · Remote Internship- Data Science Intern

Nov 2023

- Analyzed the features to understand the predictive value for breast cancer classification and diagnosis. Then, created models using six different ML algorithms and used the models to predict the cancer diagnosis as benign or malignant.
- ***Skills:*** *Python · Computer Vision · Image processing · Cascading Style Sheets (CSS) · Flask · HTML*

Ernst & Young New Delhi · On site - Tech Consulting Intern in Data Analytics Division

Aug 2023- Oct 2023

- Leveraged ML algorithms to personalize the healthcare navigation app for improved user experience.
- Analyzed data and developed a model to integrate cutting-edge AI solutions in collaboration with the development team.
- Contributed as a frontend developer in a CDAC and EY initiative for smart city development in India.
- ***Skills:*** *React.js · Node.js · GitHub · Git · Flutter*

Straive · Remote Internship- Technology Services Intern

Sep 2023- Oct 2023

- Used OpenCV in Python for manipulation detection and plagiarism checks on images, optimized key point detection with ORB and HDBSCAN, and implemented homography and RANSAC algorithms after image pre-processing.
- ***Skills:*** *OpenCV · Python · Scikit-Learn · Image Processing · Feature Extraction · NumPy · Feature Matching · Computer Vision · Matplotlib*

PROFESSIONAL COURSES & TRAINING

Biology Meets Programming: Bioinformatics: UC San Diego

- Studied computational genealogy algorithms and built a genome sequencer to apply to biology data.
- ***Skills:*** *Bioinformatics Suffix Tree, Unweighted Pair Group Method with Arithmetic Mean (UPGMA), Viterbi Algorithm, Algorithms, Python programming, Whole Genome Sequencing*

Genome Sequencing: Bioinformatics- UC San Diego

- Studied genome sequencing and its potential to usher in an era of personalized medicine with a specific use case of a deadly Staphylococcus bacterium that has acquired antibiotics resistance.
- ***Skills:*** *Algorithms, Python Programming, Whole Genome Sequencing, Dynamic Programming*

Genomic Data Science Specialization- John Hopkins University

- Studied concepts and tools used in genomic data science to understand, analyze, and interpret data from next generation sequencing experiments.
- ***Skills:*** *Bioinformatics Algorithms, Python, Algorithms On Strings, Bioinformatics, Biopython, Genomics, Statistics, Data Science, Computational Biology, Samtools, Command-Line Interface, Unix, Data Analysis, R Programming, Biostatistics, Bioconductor*

LEADERSHIP & VOLUNTEERING EXPERIENCE

- **Saathi Care-** Co founded in Sep 2023 for transforming at-home eldercare in Pune, India. Currently caregiving 45+ senior citizen families.
- **Mentored a workshop on UI/UX Design** organized by Skill Nation at VIT with over 50 attendees.
- **Marketing and Publicity Coordinator for VIT Cultural Fest**, Riviera (2024), managed promotional campaigns, secured chief guests, coordinated event discussions, and led over 120 volunteers.
- **Head of Marketing for Youth India Network**, a social media news broadcast channel that exclusively delivers video content, focusing on sports, business, education, employment, and science and technology news.
- **Project Manager for the Good Health and Well-being segment for Muskaan,** Oversaw community health and wellness initiatives & implemented innovative solutions to address healthcare challenges and promote good health and well-being
- **Write articles for the website Medium.com,** including ***'Who Deserves Innovation? Why Equitable Healthcare Is the Future', and 'From Code to Cure: Women Driving Innovation in Bioinformatics and Drug Discovery'.***